MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2025
THIRD SEMESTER
Discipline Specific Core Course (DSC B) - MG3DSCBTG202- FUNDAMENTALS OF MOLECULAR BIOLOGY
(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A - One word answer - Answer any 10 questions. Each question carries 2 marks

1. Who proposed central dogma of molecular biology? [K] / [CO2]
2. Where is prokaryotic and eukaryotic genome located in a cell? [An] /[CO1]
3. Expand SINES and LINES [U] / [CO1]
4. In which DNA repair mechanism is DNA glycosylase involved? [U] / [CO3]
5. Give two examples of chemical mutagens. [U] / [CO3]
6. Name the enzymes involved in elongation of lagging strand inDNA replication [An] / [CO2]
7. What are the componentsof an operon? [U] / [CO3]
8. Which upstream region acts as RNA Pol binding site in prokaryotic transcription? [K] / [CO2]
9. Which nucleic acids can act as the genome in a virus? [An] / [CO1]
10. Which experiment identified DNA as the 'transforming principle'? [U] / [CO1]
11. What is the function of SSB protein in DNA replication? [An] / [CO2]
12. What is the role of ribosomes in gene expression? [U] / [CO3]

(10X2= 20)

Part B- Short essay- Answer any 5 questions. Each question carries 4 marks

13. Describe 5 milestone discoveries and its impact in the field of molecular biology. [U] / [CO1]
14. Explain the differences in prokaryotic and eukaryotic DNA replication. [An] / [CO2]
15. Discuss the errors in genome that can contribute to point mutations. [U] / [CO3]
16. How is a prokaryotic genome packed inside its cell? [U] / [CO1]
17. Describe the structure of a eukaryotic gene [U] / [CO2]
18. Differentiate between positive and negative gene regulation [U] / [CO3]
19. With a neat labelled diagram, describe the structure of t-RNA. [U] / [CO3]

(5X4=20)

Part C- Essay- Answer any 1 question. Each question carries 10 marks

20.	Describe different DNA repair mechanisms in detail.	[U] / [CO3]
21.	Differentiate the DNA replication in prokaryotes and eukaryotes.	[An] / [CO2]

(1X10=10)